

Case Study 1: Student Performance Prediction

Objective

Predict exam score using:

 x_1 : Study Hours x_2 : Attendance PercentageDependent Variable: y = Exam Score

Dataset

Student	Study Hours (x_1)	Attendance (x_2)	Score (y)
1	2	70	55
2	4	75	68
3	5	80	72
4	6	85	80
5	8	90	92

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Calculation Table

x_1	x_2	y	x_1^2	x_2^2	$x_1 y$	$x_2 y$	$x_1 x_2$
2	70	55	4	4900	110	3850	140
4	75	68	16	5625	272	5100	300
5	80	72	25	6400	360	5760	400
6	85	80	36	7225	480	6800	510
8	90	92	64	8100	736	8280	720
$\Sigma 25$	400	367	145	32250	1958	29790	2070

Normal Equations

$$\sum Y = na + b_1 \sum X_1 + b_2 \sum X_2$$

$$\sum X_1 Y = a \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2$$

$$\sum X_2 Y = a \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2$$

Substituting values:

$$367 = 5a + 25b_1 + 400b_2$$

$$1958 = 25a + 145b_1 + 2070b_2$$

$$29790 = 400a + 2070b_1 + 32250b_2$$

Solving: $a = 7.52$

$b_1 = 4.82$

$b_2 = 0.55$

Regression Equation -

$$\hat{Y} = 7.52 + 4.82X_1 + 0.55X_2$$

Predicted Values and Residuals

Student	Actual (Y)	Predicted ($\hat{Y}$)	Residual ($Y - \hat{Y}$)	R^2
1	55	55.66	-0.66	0.4356
2	68	68.05	-0.05	0.0025
3	72	75.62	-3.62	13.1044
4	80	83.19	-3.19	10.1761
5	92	95.58	-3.58	12.8164

$$SSE = \sum (y - \hat{y})^2 = 36.535$$

Calculation of R^2 :

$$\text{Mean score: } \bar{y} = \frac{367}{5} = 73.4$$

$$SST = \sum (y - \bar{y})^2 = 759.20$$

$$R^2 = 1 - \frac{SSE}{SST}$$

$$= 1 - \frac{36.535}{759.20} = 0.952$$

$$R^2 = 95.2\%$$

Calculation of RMSE:

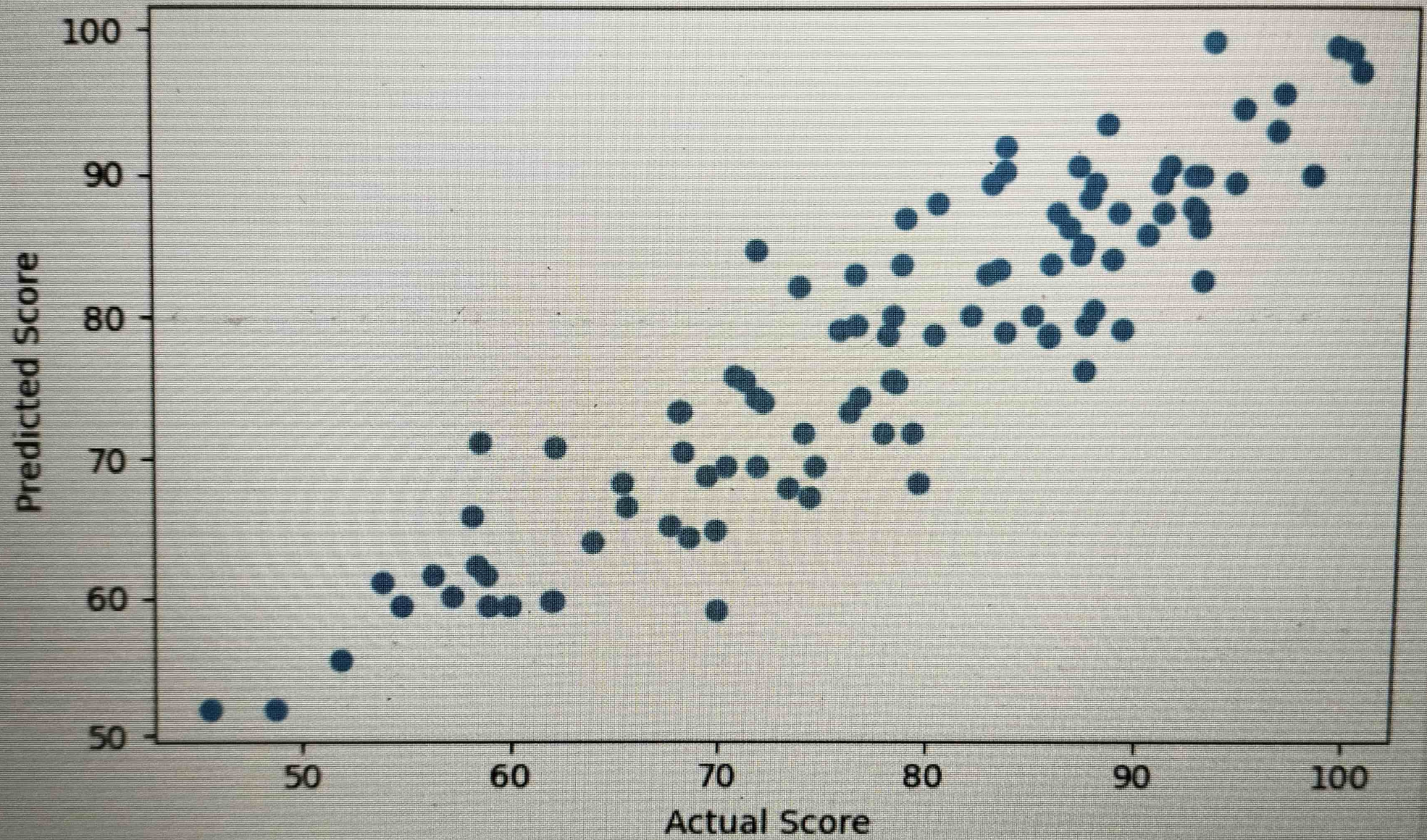
$$RMSE = \sqrt{\frac{SSE}{n}} = \sqrt{\frac{36.535}{5}}$$

$$RMSE = 2.70$$

Interpretation

- Each additional study hour increases score by approximately 4.82 marks
- Every 1% increase in attendance increases score by approximately 0.55 marks
- Study hours have a greater impact on performance.
- The model explains 95.2% of score variation.

Case 1: Actual vs Predicted Exam Scores



- Average prediction error is 2.70 marks

Key insights

- Encourage students to increase study hours.
- Maintain regular attendance.
- Study time is the strongest predictor.

Case Study 2: Retail Sales Prediction

Objective

Predict sales using:

X_1 : Advertising spend (₹ Lakhs)

X_2 : Discount (%)

Dependent Variable: Y = Sales

Dataset:

Month	Ad spend (x_1)	Discount (x_2)	Sales (y)
Jan	10	5	145
Feb	12	8	185
Mar	15	10	185
Apr	18	12	210
May	20	15	235

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Calculation Table

x_1	x_2	y	x_1^2	x_2^2	$x_1 y$	$x_2 y$	$x_1 x_2$
10	5	145	100	25	1450	725	50
12	8	165	144	64	1980	1320	96
15	10	185	225	100	2775	1850	150
18	12	210	324	144	3780	2520	216
20	15	235	400	225	4700	3525	300
Σ 75	50	940	1193	558	14685	9940	812

Normal Equations

$$940 = 5a + 75b_1 + 50b_2$$

$$14685 = 75a + 1193b_1 + 812b_2$$

$$9940 = 50a + 812b_1 + 558b_2$$

Solving $a = 80.25$

$$b_1 = 4.15$$

$$b_2 = 4.85$$

Regression Equation

$$\hat{y} = 80.25 + 4.15x_1 + 4.85x_2$$

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Month	Actual (Y)	Predicted (Ŷ)	Residual (Y - Ŷ)	R ²
Jan	145	146	-1	1
Feb	165	168.85	-3.85	14.8225
Mar	185	191	-6	36
Apr	210	213.15	-3.15	9.9225
May	235	236	-1	1

$$SSE = \sum (Y - \hat{Y})^2 = 62.745$$

Calculation of R²

Mean Sales: $\bar{Y} = \frac{940}{5} = 188$

$$SST = \sum (Y - \bar{Y})^2 = 5080$$

$$R^2 = 1 - \frac{SSE}{SST}$$

$$= 1 - \frac{62.745}{5080} = 0.988$$

$$\boxed{R^2 = 98.8\%}$$

Calculation of RMSE

$$RMSE = \sqrt{\frac{SSE}{n}}$$

$$= \sqrt{\frac{62.745}{5}} = 3.54$$

Interpretation:

- Every ₹1 lakh increase in advertising increases sales by approximately 4.15 units.
- Every 1% increase in discount increases sales by approximately 4.85 units.
- Discount has a slightly stronger effect on sales.
- The model explains 98.8% of sales variation.
- Average prediction error is 3.54 units

Key Insights:

- Both advertising and discounts positively affect sales.
- Discounts generate higher sales growth.
- Excessive discounts may reduce profits.
- A balanced marketing strategy is recommended.

Case 2: Actual vs Predicted Sales

